

Exercise 04: SQL DML: SFW blocks

In the last exercise you created the `Students`, `Lecturers`, `Events` and `Student_in_Event` tables with the known attributes and keys. Those definitions are provided as a script to this exercise.

Task 1: Simple selects

- Create a list of all lecturers who have their office in wing D.
- Create a list of matriculation numbers of students who have not yet received a grade in the summer semester 2018.
- Create a list with the headings `Student` and `Age`, in which the names of the students and their ages are listed. Prevent the output of students who are younger than 20 and older than 40. The required T-SQL commands are `current_timestamp` and `datediff`.

Check your solutions using the solution of the last exercise sheet or enter suitable test data yourself.

Task 2 : Join and cross product

- * Create a room plan for your fellow students for the summer semester 2018 that shows who is in which event in which room. In doing so, do not output any unnecessary columns and use meaningful identifiers for the output columns.
- Create a table in which the name of a student is in the first column and the name of an younger student is in the second column with the label `„is older than“`. Generate all possible combinations of rows with this relationship and sort the output based on the first column, with the oldest student displayed first.
- * Create a table with only one column, which you call `Text`. In each row, there is a text based on the following pattern: *“Eva participated in the lecture beach volleyball during the summer semester 2018 and obtained the grade 4.0. Congratulations!”* Only output rows for the winter semester 2017, the summer semester 2017 and the summer semester 2018. Adjust the text for each line accordingly and also address the grade (for example *“did not pass”* or *“no grade so far”*).

Kommentiert [LA1]: Ausgangstext: Erzeugen Sie eine Tabelle bei der in der ersten Spalte der Name eines Studenten angezeigt wird und in der zweiten Spalte mit der Bezeichnung „...ist älter als“ der Name eines jeweils älteren Studenten angezeigt wird.
Macht das Sinn? „is older than“ ist Spalte 2? Bitte prüfen ☺

Task 3 : Quantifiers and set operations

- * Create a table which shows the best grade a lecturer has ever given for a specific event across all semesters.
Additional task: how do you handle `null` values in grades (look at the `isnull` function).
- Extend the above task to also indicate the worst grade given.
- Additional task: how can you easily check whether `Student`, `Event` and `Semester` fulfil the key property in `Student_in_Event`? How can you output only the tuples that violate the key property?